# Anti-Involution and Global Price Pressure: China's New Supply-Side Reform and the Spillovers of Green Industrial Overcapacity

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## 1 Introduction and the Policy Engine

China accounts for more than 70% of global electric-vehicle production (IEA, 2025a) and holds more than 80% of manufacturing capacity across key solar PV supply-chain stages. A single policy apparatus produced both positions. The state channels capital and labor toward centrally designated strategic industries, and in doing so it has created globally competitive firms. It has also generated chronic overcapacity and persistent domestic price wars, both of which now spill into trade friction.

I argue that the anti-involution campaign central authorities announced in 2025 is a supply-side reform *within* the industrial-policy state, not a retreat from it. Anti-involution targets the destructive price competition that follows from subsidized overentry, but it does not address the demand-side imbalance that makes overentry profitable in the first place. Without complementary demand-side reform, anti-involution may discipline the worst domestic price wars while the global economy continues to absorb Chinese overcapacity as downward pressure on traded-goods prices.

The sequence begins with central-government target-setting. Programs such as Made in China 2025 and the designation of the "new three" (electric vehicles, lithium-ion batteries, and solar PV) as national champions signal where state resources will flow. Local governments then compete to attract investment in these sectors. Fang et al. (2025b) find that this competition operates through a tournament mechanism: provincial and municipal officials, whose career advancement depends on measurable economic performance, offer subsidized land and cheap credit to firms willing to locate within their jurisdictions. Tax holidays and purpose-built industrial parks round out the package. The central government sets objectives; local governments compete on implementation, producing parallel investment across dozens of provinces that each build capacity in the same industries.

Soft budget constraints prevent market discipline from correcting the resulting oversupply. In a hard-budget-constraint environment, losses force exit and bankruptcy clears the surplus. In China, state-controlled banks and local government financing vehicles absorb losses instead, because lenders face political incentives to extend credit rather than recognize write-downs. The clearing mechanism is blunted. Firms that would otherwise fail keep operating, and new entrants keep arriving, because the cost of entry is socialized while the political reward for attracting investment remains private to the local official who secured it. The result is predictable: subsidized expansion without proportional gains in efficiency. Branstetter and Li (2022) find exactly this pattern for Made in China 2025 targets, and Branstetter et al.

(2023) confirm it for Chinese industrial subsidies more broadly. Subsidized firms invest more. They do not consistently become more productive.

The final link is the gap between output and domestic absorption. China's household consumption share of GDP remains low by international standards. The growth model systematically channels income toward investment, and households respond to weak social insurance by saving. High savings rates, sustained in part by weak social insurance and the post-2021 property adjustment, compress domestic purchasing power further. The implication for industrial policy is direct: every unit of capacity added to a targeted sector must find a buyer either at home or abroad. When domestic demand is structurally insufficient, the surplus generates price wars at home and export pressure abroad. Both outcomes trace to the same structural imbalance.

Policy diffusion reinforces the pattern. Fang et al. (2025a) document that local governments systematically imitate peer jurisdictions' industrial policies, duplicating capacity across hundreds of cities. The resulting subsidies have trade consequences: Rotunno and Ruta (2024) find that Chinese industrial subsidies are associated with increased export volumes in subsidized sectors.

If this account is right, overcapacity should concentrate in centrally targeted sectors. The "new three" provide the test case.

## 2 Overcapacity in the "New Three"

Table 1 summarizes capacity pressure across China's three flagship clean-technology sectors. In each, installed capacity exceeds plausible demand and output prices have fallen sharply.

**Solar PV.** Solar module prices collapsed between 2023 and 2025. The arithmetic behind the collapse is straightforward: China's solar photovoltaic manufacturing capacity exceeds 1 TW per year, while global installations reached roughly 600 GW in 2024 (IEA, 2025b). Even if every panel installed anywhere in the world came from a Chinese factory, nearly half of Chinese capacity would sit idle. Chinese firms account for more than 80% of global output at every major stage of the supply chain, from polysilicon and wafer production through cell and module assembly. Cost advantages alone do not explain this dominance. The OECD's MAGIC database identifies solar as the most heavily subsidized sector it tracks: government support to solar firms averaged 3.2% of revenue, compared with 0.9% across all sectors in the database (OECD, 2026, 2025). The subsidy wedge explains why capacity continued to expand even as prices fell below the point where unsubsidized entry would have stopped.

**Lithium-ion batteries.** China hosts about 85% of global battery cell manufacturing capacity (USCC, 2025; IEA, 2025a). Batteries differ from solar PV in one important respect: they are an intermediate good whose demand depends on downstream adoption of EVs and grid storage, so the overcapacity calculation rests heavily on expectations. The investment thesis is that global battery demand will more than triple by 2030, and firms building capacity now will capture that future market. The thesis may prove correct in the long run, but it does not solve the near-term problem. Battery pack prices in China fell about 30% in 2024, compared with 10–15% in Europe and the United States (IEA, 2025a). If the price decline reflected a global technology-driven cost curve, the drop would be roughly uniform across markets. The fact that Chinese prices fell roughly twice as fast as Western ones points to a domestic glut, not shared progress down the learning curve.

**Electric vehicles.** The EV sector is different again, and politically the most visible case. Because automobiles are a consumer good with strong local-employment ties, provincial governments have direct stakes in keeping regional automakers alive. When national purchase subsidies ended in early 2023, Chinese EV makers launched successive rounds of price cuts, each intended to gain market share at rivals' expense. The price war has not stopped. NBS data show that automobile manufacturing PPI fell 2.8% in 2025, slightly worse than the 2.6% decline for industry as a whole (NBS, 2026). Global EV sales exceeded 17 million units in 2024, but Chinese production capacity is widely reported to exceed even the combined total of domestic absorption and exports (IEA, 2025a; Mayr et al., 2026). The European Parliament study documents a rising share of loss-making firms in the automobile sector (Mayr et al., 2026). In a competitive market, persistent losses lead to exit. In China's EV sector, the soft budget constraint described in Section 1 keeps loss-making firms alive, and the price floor that exit would establish never materializes.

| Sector | China-side indicator | Demand benchmark | Price/margin symptom |
|---|---|---|---|
| Solar PV | Mfg. capacity $>$1 TW/yr | $\sim$600 GW installed (2024) | Sharp price decline |
| Li-ion batteries | $\sim$85% of global mfg. capacity | Below installed capacity | Pack prices fell $\sim$30% in China (2024) |
| EVs | Capacity exceeds demand | Global sales $>$17M (2024) | Auto mfg. PPI $-$2.8% (2025) |

Table 1: Descriptive indicators of capacity pressure in China's "new three" sectors. Solar: IEA (2025b); OECD (2026). Batteries: USCC (2025); IEA (2025a). EVs: IEA (2025a); NBS (2026).

The definition of overcapacity is contested. Beijing maintains that current capacity is forward-looking: global green demand will eventually absorb what today appears surplus. The OECD and IMF treat the gap as structural, rooted in the subsidy architecture described in Section 1 (IMF, 2026; OECD, 2025). I follow the structural interpretation. The soft budget constraint means that capacity investment reflects the incentives of local officials and state lenders, not any firm's assessment of future demand. The "forward-looking" label implies a market rationality that the tournament mechanism systematically overrides. And even on a forward-looking reading, financing the interim capacity gap imposes real costs on credit allocation and on competing producers forced to exit.

The mechanisms differ across sectors. Solar overcapacity is driven by raw arithmetic; the battery glut reflects speculative overinvestment in anticipated demand. EVs add a political dimension because local governments protect employment-intensive automakers. What the sectors share is a fiscal and financial architecture that prevents the surplus from self-correcting. Margins have compressed to the point where firm survival depends on continued state support. Beijing has noticed.

# 3 Anti-Involution as Domestic Response

## The policy and its instruments

"Involution-style competition" (*neijuan shi jingzheng*) describes a pattern in which firms compete on price and scale alone, compressing margins without generating productivity gains. No firm gains a durable advantage because competitors match each price cut: the equilibrium is a race to the bottom in which every participant loses margin and no one gains market power. The result is value destruction rather than creative destruction. Anti-involution is Beijing's label for this pathology, and its policy response.

Central authorities moved to address involution on two tracks. The first is administrative. In July 2025, a State Council executive meeting on the new-energy vehicle industry called for curbing "irrational competition" and announced strengthened cost surveys and price monitoring across the NEV supply chain (State Council, 2025a). In January 2025, the NDRC issued guidelines for building a unified national market, targeting local protectionism and regional market barriers that fragment domestic demand and enable duplicated capacity (State Council, 2025b). Specific instruments include tighter safety and quality standards (revised EV standards take effect in 2026) and merger facilitation, with local-government performance evaluation to shift from output quantity toward quality metrics.

The second track is more market-oriented in design. The revised Anti-Unfair Competition Law, adopted in June 2025, prohibits platform operators from forcing merchants to sell below cost and restricts large enterprises from abusing their bargaining position against smaller suppliers. A concurrent overhaul of the Pricing Law aims to establish legal liability for predatory pricing and other "involutionary" practices. These provisions move beyond administrative guidance and toward enforceable legal rules. I take the legal track seriously. If enforced, it would introduce a rule-of-law floor beneath pricing behavior that the administrative instruments alone cannot provide. But enforcement must run through local courts and agencies that answer to the same officials whose investment incentives drive overcapacity in the first place. The gap between legislative intent and local implementation has been a recurring feature of Chinese economic governance.

## The MITI analogy and its limits

A serious counterargument holds that anti-involution represents genuine policy learning: a maturation from quantity-maximizing industrial policy toward quality-focused upgrading. The analogy to Japan's MITI is tempting. During the 1960s and 1970s, MITI used administrative guidance to curb "excessive competition" (*kato kyoso*) in steel and automobiles, facilitating consolidation that contributed to subsequent technological upgrading (Johnson, 1982).

But did MITI succeed because of administrative coordination alone? Japan's story involved several things at once. Keiretsu organization concentrated decision-making, an open American export market absorbed output. The financial system channeled household savings toward industry at low cost. All of these mattered. I emphasize one factor that gets less attention in the standard account. Japan's industrial consolidation coincided with one of the fastest sustained increases in household living standards in economic history, and rising domestic incomes absorbed much of what industry produced. MITI could consolidate supply because demand was growing fast enough to make the consolidated firms profitable. China faces the opposite. Household consumption remains around 39–40% of GDP, far below advanced-economy benchmarks (World Bank, 2025), and the household savings rate sits at approximately 34%, against a global average of roughly 24% (Gunter et al., 2025). As Klein and Pettis (2020) argue, financial repression and the systematic channeling of income toward investment are structural features of the growth model. Consumption is suppressed by design.

A fair objection is that this framing is too narrowly domestic. China is not betting on Chinese demand alone; it is betting on global green-transition demand. If the world triples its installed solar capacity and electrifies its vehicle fleet on the timeline climate policy requires, Chinese firms with today's surplus capacity will be well-positioned to supply that demand. The bet may prove right over a decade or more. But it does not solve the near-term problem: the interim costs of financing idle capacity and sustaining loss-making firms fall on Chinese taxpayers now, while the trade retaliation those exports provoke falls on Chinese exporters. And the bet itself is partly self-defeating, because the trade barriers Chinese overcapacity provokes are precisely what could prevent the global market from reaching the scale that would justify the capacity.

Anti-involution may slow the price war among EV manufacturers. It cannot create the domestic buyers those manufacturers need. The textbook prescription would be to raise the household share of income through fiscal transfers and expanded social insurance, coupled with a relaxation of financial repression. None of these appear in the anti-involution policy toolkit.

Scale compounds the difficulty. MITI coordinated a handful of firms in a few sectors; China's overcapacity spans hundreds of cities and thousands of firms (Gunter et al., 2025). The fiscal architecture of the state works against compliance. Local governments depend on land-transfer fees and industrial-project revenue, and the tournament mechanism documented in Section 1 persists because these fiscal incentives persist. Anti-involution creates a principal-agent problem: Beijing issues the directive, but the agents who must implement it are the same local officials whose budgets depend on the investment the directive is trying to restrain.

I therefore read anti-involution as primarily a correction within the existing system, blending administrative discipline with emerging legal instruments but leaving the demand side untouched. It may reduce the worst price wars and, if the legal reforms are enforced, establish a floor beneath pricing behavior. But production capacity that exceeds domestic demand, sustained by a growth model that favors investment over consumption, is left intact. Whatever the domestic market cannot absorb will find its way abroad.

# 4 Global Spillover: Downward Price Pressure

## China Shock 2.0

When Autor et al. (2013) documented the original China Shock, the story was straightforward: WTO accession accelerated a trade shock rooted in comparative advantage in labor-intensive manufacturing, and American workers in competing industries paid the price. Two decades later, the shock has returned in a different form. What China now exports in volume is capital-intensive green technology: the "new three" of EVs, batteries, and solar PV. The goods are different. The labor-market disruption in importing countries looks familiar.

The continuity is worth stressing because the underlying mechanism is different enough to obscure it. China possesses genuine endowments in engineering talent and scale economies, but the policy apparatus described earlier amplified these advantages through directed capital allocation and protected domestic markets. The current shock layers policy-amplified capacity expansion on top of real cost advantages. The welfare calculus is more complex because the goods carry positive externalities: cheap solar panels and batteries slow climate change. The original China Shock asked whether cheap manufactures justified displaced workers. This one asks the same question with an additional complication.

## Strategic entry versus residual exports

Not all Chinese green-tech exports reflect the same dynamic, and the distinction matters for policy. Some firms built export-oriented capacity strategically, planned before domestic margins collapsed. BYD's plants in Thailand and Hungary are examples: deliberate market entry into new geographies, the kind of outward FDI that any capital-exporting economy would produce. Others are redirecting output abroad because the domestic price war left them unable to sell at home. This second category is closer to losstolerant export competition, where firms accept margins below long-run average cost and the soft budget constraint extends their survival horizon well beyond what private capital markets would finance.

Trade statistics do not distinguish between strategic entry and residual dumping. In practice, firms optimize across domestic and export markets simultaneously, and the same firm may be strategic in one market and residual in another. But the structural point holds regardless: when aggregate capacity exceeds domestic absorption, the export market functions as a pressure valve, and the pressure is downward on prices.

## Evidence of downward price pressure

Chinese export volumes in clean-technology sectors have risen sharply since 2022 while unit values have fallen. The USCC documents this pattern across several sectors (USCC, 2025). Rotunno and Ruta (2024) provide econometric evidence that the relationship is systematic: Chinese industrial subsidies are associated with increased export volumes and depressed export prices in subsidized sectors. The direction of the effect is what the earlier sections of this essay would predict. Subsidies expand capacity beyond what domestic demand can absorb, and the surplus enters world markets at prices that unsubsidized competitors find difficult to match.

A note on terminology. I avoid "exported deflation" because the phrase implies a general price-level effect. What is occurring is a decline in the relative price of clean-technology goods in world markets: a terms-of-trade improvement for importing countries but a deterioration for China and for competing producers elsewhere. The policy problem is sectoral adjustment, not macroeconomic deflation management.

## The green-public-goods counterargument

The strongest objection to treating overcapacity as a problem runs as follows. Cheap Chinese solar panels and batteries accelerate global decarbonization, particularly in developing countries that cannot afford unsubsidized clean technology. On standard trade-theory grounds, a subsidy on goods with large positive externalities may be globally welfare-improving. For many countries in Sub-Saharan Africa and South Asia, affordable solar modules are the difference between decarbonized electrification and continued fossil-fuel reliance.

Yet the welfare case is less clean than this framing suggests. Adjustment costs fall unevenly: workers in European solar-module assembly or American EV-component manufacturing bear concentrated losses while benefits diffuse across consumers, replicating the distributional pattern that made the original China Shock politically corrosive. Dependence on a single subsidized source also creates supply-chain vulnerability at upstream chokepoints such as polysilicon refining. The trade-policy responses already deployed are fragmenting the very markets through which welfare gains would flow. The United States imposed a 100% tariff on Chinese EVs in 2024 (USTR, 2024); the European Union adopted countervailing duties ranging from 17.0% for BYD to 35.3% for SAIC (European Commission, 2024). In a first-best world, importing countries would accept subsidized goods and compensate displaced workers through domestic transfers. No major economy has chosen this path. And yet the welfare benefit of subsidized overcapacity depends on open markets, which the overcapacity itself is closing. Subsidies meant to accelerate decarbonization are accelerating trade fragmentation instead.

This leaves the system in an unstable position. Domestic demand cannot absorb the capacity, and export markets are fragmenting under tariff pressure. Something has to give.

# 5 Tensions and Outlook

Anti-involution targets the wrong end of the problem. It can discipline the price competition that drives margins below zero. Insufficient domestic demand stays untouched, and that is the deeper issue. Raising quality thresholds and facilitating consolidation does not create buyers.

The most likely trajectory is a muddled middle on the domestic front. Solar manufacturing may see partial rationalization as small firms exit, but the EV sector is harder to consolidate because local governments hold stakes in regional automakers. No official wants to close a factory that employs constituents and generates tax revenue, even if the factory cannot turn a profit.

The more interesting adjustment channel may lie abroad. Chinese EV supply-chain firms invested more overseas than domestically for the first time in 2024 (IEA, 2025a), and industry estimates project overseas manufacturing capacity belonging to Chinese automakers will roughly double by 2026. BYD alone is building or planning plants in Thailand and Hungary, with further investments announced in Turkey and Brazil. This outward FDI is partly tariff-driven: firms localize production to circumvent the trade barriers that finished-goods exports now face. But it also represents a structural shift. If Chinese firms relocate a portion of their capacity to Southeast Asia and Latin America, they may endogenously relieve some of the domestic overcapacity pressure that anti-involution was designed to address. The Global South remains broadly open to Chinese clean-technology imports; over half of the growth in Chinese EV exports in 2025 came from non-OECD markets. For these countries, affordable Chinese solar modules and batteries are a path to electrification that no Western supplier currently matches on price.

This outward shift changes the nature of the problem rather than solving it. The overcapacity is partially externalized through relocation rather than absorbed through domestic demand growth. The distributional conflict between Chinese producers and Western competitors does not disappear; it migrates to new geographies. And the fiscal incentives within China that drive local officials to build capacity remain intact. Outward FDI relieves pressure. It does not fix the valve.

Genuine structural reform would require raising the household share of national income through expanded social insurance and fiscal restructuring away from land-sale dependence, coupled with a relaxation of financial repression. Each is politically difficult because each redistributes income away from the state sector. Anti-involution asks none of these questions.

Yet the campaign does reveal something about the state's capacity for self-correction. The same apparatus that created globally competitive green-manufacturing sectors now attempts to regulate the overcapacity it produced. The industrial-policy state is turning its own tools inward. Whether the correction holds depends less on administrative will than on whether domestic consolidation and outward FDI together can absorb the surplus that domestic consumption cannot. The precedents for demand-side reform are not encouraging: China has discussed rebalancing toward consumption since at least the 12th Five-Year Plan (2011–2015), and the household consumption share of GDP has barely moved.

What's visible is a Chinese industrial sector adapting to the constraints it faces, finding new outlets for overcapacity rather than addressing its origins.

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